Hardware Documentation

PARMCO Project — Checkpoint 3
AI-Generated Documentation
Subsystem: Motor, Isolation, IR Sensor & Encoder

Table of Contents

1. System Overview 1
2. Bill of Materials 1
3. GPIO and I2C Assignment 2
4. Pin-to-Pin Assembly Guide 3
5. Assembly and Bring-Up Instructions 3
6. How It Works (Functional Analysis) 5
6.1 Power and isolation logic (Quiet Start) 5
6.2 Bipolar drive sequence 5
6.3 Speed and step timing 5
6.4 Current, power and decoupling 6
6.5 IR sensor and RPM measurement 6
6.6 AS5600 magnetic encoder 6
Schematic: Isolation, Drive, IR Sensor & Encoder 7

1. System Overview

The PARMCO hardware architecture safely interfaces a 3.3V Raspberry Pi 4 with a 12V Nema17 bipolar stepper motor (XY42STH34-0354A: 1.8° per step, rated 12 V and 0.35 A per phase). A 4N25 optocoupler is the on/off enable to the H-bridge: the Pi's GPIO enable signal only drives the optocoupler LED, so the motor stays disabled during the Pi's boot sequence ("Quiet Start"). An SN754410NE Quad Half-H Bridge translates the logic-level half-step sequence from four GPIO pins into the high-current phases that drive the two motor coils.

An **IR sensor** detects the two propeller blades and feeds one digital pulse per blade to a Pi GPIO input, so the software can report propeller RPM. Checkpoint 3 adds an **AS5600 magnetic encoder** on the Pi's I2C bus. A magnet on the motor shaft turns above the AS5600, which reports the shaft angle as a 12-bit number; the software uses it for calibrate zero, set to zero, angle targets and the encoder-derived RPM. The phone link uses the Pi's on-board Bluetooth radio and needs no additional circuitry.

Changes from Checkpoint 2: the AS5600 magnetic encoder and its magnet are added (parts list, I2C assignment, pin table, schematic); the assembly and bring-up instructions cover the encoder; the "how it works" analysis has a new encoder section (6.6). The motor drive, 4N25 enable, pull-down, decoupling and IR sensor circuits are unchanged.

2. Bill of Materials

Component	Purpose	Specification
Raspberry Pi 4	Main processing unit. Runs the C/ARM assembly control program, reads the IR sensor and encoder, hosts the Bluetooth server.	3.3V Logic via Protector HAT; on-board Bluetooth
Nema17 Stepper Motor	Primary actuator, modified with 3D-printed stand.	XY42STH34-0354A, 1.8° Step (Bipolar), 12 V, 0.35 A per phase

Component	Purpose	Specification
SN754410NE	Quad Half-H Bridge. Drives current to motor coils (two H-bridges).	Min 1A Output Capacity; VCC1 logic 5 V, VCC2 motor 12 V; internal output clamp diodes
4N25 Optocoupler	Enable path to the H-bridge; keeps the motor disabled at boot.	NPN Phototransistor, 6-pin DIP (pin 3 NC, pin 6 base left unconnected)
Resistor R1	Limits GPIO17 current through the 4N25 LED.	320 ohm (about 6.6 mA LED current at 3.3 V)
Resistor R2	Pull-down on the 4N25 emitter / SN754410NE enable node (pins 1 and 9): holds the enables low whenever the 4N25 is off.	10 kohm, 4N25 pin 4 to ground
Capacitor C1	Decoupling for the 12V motor rail (VCC2).	100 µF, rated above 12 V, polarized (+ to 12V)
Capacitor C2	Decoupling for the 5V logic rail (VCC1).	0.1 µF
IR Sensor module	Detects propeller blades: 2 pulses per revolution, used for RPM.	Digital output, HIGH while a blade is detected; powered from the Pi 3.3V pin
AS5600 magnetic encoder module	Measures the shaft angle (0-360°) over I2C; mounted on the class 3D-printed motor stand.	12-bit (4096 counts per revolution), I2C address 0x36, powered from the Pi 3.3V pin; DIR tied to ground
Encoder magnet	Turns with the motor shaft above the AS5600.	Diametrically magnetized, centered on the shaft axis
DC Power Supply	Provides dual-rail power to the logic and motor circuits.	CH1: 5V (0.2A), CH2: 12V (1A)
2-Blade Propeller	Rotational visual indicator (trimmed for clearance); one blade marked as the reference blade.	Plastic, Friction Fit
Breadboard and jumper wires	Common ground rail and circuit interconnect.	Solderless breadboard
Android phone	Bluetooth RFCOMM client (not part of the circuit).	Connects to the Pi's on-board Bluetooth; no extra wiring

3. GPIO and I2C Assignment

GPIO numbers are the BCM numbers used by the class RP4 library and by the Checkpoint 3 code. The I2C pins are used through the Linux I2C driver (/dev/i2c-1), not through the class library.

Signal	BCM GPIO	Header Pin	Direction	Connects to
Motor enable	GPIO17	11	Output	4N25 Pin 1 (Anode) via R1 (320 ohm)
Coil input 1A	GPIO22	15	Output	SN754410NE Pin 2
Coil input 2A	GPIO23	16	Output	SN754410NE Pin 7
Coil input 3A	GPIO24	18	Output	SN754410NE Pin 10
Coil input 4A	GPIO25	22	Output	SN754410NE Pin 15
IR sensor pulse	GPIO27	13	Input	IR Sensor OUT
I2C data (SDA)	GPIO2	3	Bidirectional	AS5600 SDA
I2C clock (SCL)	GPIO3	5	Output (clock)	AS5600 SCL
3.3V supply	—	1	Power	IR Sensor VCC and AS5600 VCC

Signal	BCM GPIO	Header Pin	Direction	Connects to
Ground	—	6	Ground	Common Ground Rail

4. Pin-to-Pin Assembly Guide

Source Component	Source Pin / Rail	Destination Component	Destination Pin
DC Power Supply	12V (VCC2)	SN754410NE	Pin 8 (VCC2 - Motor Power)
DC Power Supply	5.0V (VCC1)	SN754410NE	Pin 16 (VCC1 - Logic Power)
DC Power Supply	Ground (-)	Breadboard Rail	Common Ground
Raspberry Pi 4	Ground (Pin 6)	Breadboard Rail	Common Ground
Raspberry Pi 4	GPIO17 Output (Enable), Pin 11	4N25 Optocoupler	Pin 1 (Anode) via R1 (320 ohm)
4N25 Optocoupler	Pin 2 (Cathode)	Breadboard Rail	Common Ground
4N25 Optocoupler	Pin 5 (Collector)	DC Power Supply	5.0V (VCC1)
4N25 Optocoupler	Pin 4 (Emitter)	SN754410NE	Pins 1 & 9 (Enables)
Raspberry Pi 4	GPIO22, 23, 24, 25 (Step Seq)	SN754410NE	Pins 2, 7, 10, 15 (1A, 2A, 3A, 4A) respectively
SN754410NE	Pins 1 & 9 (enable node, 4N25 Pin 4)	Resistor R2 (10 kohm)	Breadboard Rail (Common Ground)
SN754410NE	Pins 4, 5, 12, 13	Breadboard Rail	Common Ground (Heatsink)
DC Power Supply	12V (VCC2) rail	Capacitor C1 (100 µF)	+ to 12V, - to Common Ground
DC Power Supply	5.0V (VCC1) rail	Capacitor C2 (0.1 µF)	One lead to 5V, one lead to Common Ground
SN754410NE	Pins 3 & 6 (1Y, 2Y)	Nema17 Motor	Coil A Pair
SN754410NE	Pins 11 & 14 (3Y, 4Y)	Nema17 Motor	Coil B Pair
Raspberry Pi 4	3.3V (Pin 1)	IR Sensor	VCC
IR Sensor	GND	Breadboard Rail	Common Ground
IR Sensor	OUT	Raspberry Pi 4	GPIO27 (Pin 13), input
Raspberry Pi 4	3.3V pin (same 3.3V net as the IR sensor)	AS5600 Encoder	VCC
AS5600 Encoder	GND	Breadboard Rail	Common Ground
Raspberry Pi 4	GPIO2 / SDA (Pin 3)	AS5600 Encoder	SDA
Raspberry Pi 4	GPIO3 / SCL (Pin 5)	AS5600 Encoder	SCL
AS5600 Encoder	DIR	Breadboard Rail	Common Ground (angle increases clockwise)

5. Assembly and Bring-Up Instructions

Wear eye protection whenever a propeller is spinning. Fit the propeller only after the quiet start and the stop command have been confirmed. Keep the DC supply outputs off and the RP4 unplugged until step 11.

Build (all power off)

1. Attach the Nema17 to the 3D-printed stand. Identify the two coil pairs by continuity with a meter (a coil pair reads a low resistance; wires from different pairs read open).
2. Place the SN754410NE and 4N25 on the breadboard, observing the pin-1 notch/dot of each. Connect the breadboard ground rail to the DC supply ground (-), RP4 Pin 6, and SN754410NE pins 4, 5, 12 and 13.
3. Connect supply CH2 (12V) to SN754410NE pin 8 and CH1 (5V) to pin 16. Neither supply pin may connect to any RP4 pin.
4. Place decoupling capacitor C1 (100 µF, positive lead to the 12V rail) between the 12V rail and the ground rail, and C2 (0.1 µF) between the 5V rail and the ground rail, close to SN754410NE pins 8 and 16.
5. Wire the 4N25: RP4 GPIO17 (Pin 11) through resistor R1 (320 ohm) to pin 1; pin 2 to ground; pin 5 to the 5V (VCC1) rail; pin 4 to SN754410NE pins 1 and 9. Connect pull-down resistor R2 (10 kohm) from pin 4 to the ground rail. Leave 4N25 pins 3 and 6 unconnected.
6. Wire GPIO22, 23, 24, 25 (Pins 15, 16, 18, 22) to SN754410NE pins 2, 7, 10, 15.
7. Wire coil A to SN754410NE pins 3 and 6, and coil B to pins 11 and 14.
8. Wire the IR sensor: VCC to RP4 Pin 1 (3.3V), GND to the ground rail, OUT to GPIO27 (Pin 13). Position the sensor so each blade passes through its detection zone (two pulses per revolution).
9. Wire the AS5600 encoder: VCC to a 3.3V pin of the RP4 (never 5V), GND to the ground rail, SDA to RP4 Pin 3 (GPIO2), SCL to RP4 Pin 5 (GPIO3), and DIR to the ground rail. Leave the OUT and PGO pins unconnected. Fit the magnet on the motor shaft and mount the AS5600 on the motor stand so the chip sits directly under the magnet's center.
10. Pre-power check: confirm ground continuity (RP4, supply, pins 4/5/12/13, IR sensor, AS5600); confirm VCC1 and VCC2 trace only to the DC supply; confirm no 5V or 12V wire touches any RP4 pin and that the AS5600 VCC is on a 3.3V pin; confirm the polarity of C1 and that R2 is in place; with a meter check that the 3.3V/GND/OUT wires of the IR sensor and the VCC/GND/SDA/SCL wires of the AS5600 are not swapped.

Power-up and verification

11. Set CH1 to 5.0 V and CH2 to 12.0 V (current limits as the instructor directs), then turn the outputs on with the RP4 still unplugged. VCC2 current must be below 0.05 A and the shaft must turn freely by hand. If the shaft locks or the current is 0.5 A or more, the enable wiring is wrong: power down.
12. Plug in the RP4 and watch the shaft and VCC2 current for the whole boot: no movement, current below 0.05 A, shaft free afterwards (Quiet Start).
13. Make sure I2C is enabled on the RP4 and that the AS5600 answers at address 0x36 (for example with `i2cdetect -y 1`). With the motor disabled, turn the propeller by hand: the encoder angle must change smoothly through 0 to 360° and increase when the propeller turns clockwise (if it decreases, the software option ENCODER_INVERT_DIR must be set to 1).
14. Start the control program. Enable the motor from the app: VCC2 current rises (roughly 0.3 A with one coil energized and 0.6-0.7 A with two, depending on the active phase) and the shaft holds. Disable: current returns below 0.05 A.
15. Run one revolution each way and compare the stop position with the start mark (one revolution is 400 half steps). Then run continuously at a moderate rate and compare the IR RPM and the encoder RPM in the app with the commanded RPM (steps per second x 0.15, see section 6.3).
16. Power down: disable the motor, shut down the RP4, then turn off the DC supply outputs.

6. How It Works (Functional Analysis)

6.1 Power and isolation logic (Quiet Start)

The SN754410NE drives current only while its enable inputs (pins 1 and 9) are high. Those inputs are fed by the emitter of the 4N25 phototransistor, whose collector is tied to the 5V logic rail (VCC1), and they are held to ground by the 10 kohm pull-down resistor R2. The RP4 GPIO17 pin drives only the 4N25 LED, through R1 (320 ohm) to ground. During Pi boot the GPIO pins float or are pulled low, so the LED is off and the phototransistor is open. R2 then holds the enable inputs at ground, far below the logic-low threshold, so the H-bridge outputs stay off and zero current is delivered to the motor coils, regardless of the 12V VCC2 state or the state of the four coil inputs. When the program sets GPIO17 high, about 6.6 mA ((3.3 V - 1.2 V) / 320 ohm) flows in the LED and the phototransistor conducts. Its photocurrent (about 1 mA or more at the 4N25's minimum 20% transfer ratio) exceeds the 0.5 mA that R2 draws at 5 V, so the enable inputs rise to roughly 5 V minus the transistor's saturation voltage, well above the logic-high threshold of the SN754410NE.

The 4N25 gives a signal path in which the RP4 pin never connects to the 5V or 12V rails, and the enable is on/off only (it is never PWM'd). The grounds of the RP4, the supply and the H-bridge are common (required, because the 3.3V step inputs need a shared reference), so this is optical coupling of the enable signal, not full galvanic isolation of the two grounds.

6.2 Bipolar drive sequence

Each coil of the bipolar Nema17 is connected between the two outputs of one H-bridge: coil A between 1Y and 2Y (pins 3 and 6), coil B between 3Y and 4Y (pins 11 and 14). An input driven high pulls its output to VCC2, an input driven low pulls it to ground. For coil A: 1A high and 2A low gives current in the + direction (A+); 1A low and 2A high gives A-; both inputs equal gives no current. Coil B behaves the same with 3A and 4A. The program steps through the 8-entry half-step table below, one entry per half step. Bit 3 of each table byte drives 1A (GPIO22), bit 2 drives 2A (GPIO23), bit 1 drives 3A (GPIO24) and bit 0 drives 4A (GPIO25). The table index advances by one for one direction and goes back by one for the other, wrapping from 7 to 0 and 0 to 7. One table pass is 8 half steps; 400 half steps (0.9° each) make one revolution.

Index	1A 2A 3A 4A	Coil A	Coil B	Byte
0	1 0 0 0	A+	off	0x08
1	1 1 0 0	off	off	0x0C
2	0 1 0 0	A-	off	0x04
3	0 1 1 0	A-	B+	0x06
4	0 0 1 0	off	B+	0x02
5	0 0 1 1	off	off	0x03
6	0 0 0 1	off	B-	0x01
7	1 0 0 1	A+	B-	0x09

```
// Bit mapping: [3]1A, [2]2A, [1]3A, [0]4A
// Sequence: 1000, 1100, 0100, 0110, 0010, 0011, 0001, 1001
half_step_table: .byte 0x08, 0x0C, 0x04, 0x06, 0x02, 0x03, 0x01, 0x09
```

6.3 Speed and step timing

In half-step mode, RPM = steps per second x 60 / 400 = steps per second x 0.15. The program's default of 20 RPM is 133.3 steps/s; 100 RPM is 666.7 steps/s, or 1.5 ms between steps.

6.4 Current, power and decoupling

At rated 12 V and 0.35 A per phase, a single energized coil draws about 0.3-0.35 A (less the H-bridge saturation drop) and a two-coil phase about 0.6-0.7 A. This is within the 1 A supply channel and the 1 A per-channel rating of the SN754410NE. VCC1 (5V) supplies the SN754410NE logic and the 4N25 collector current, far below the 0.2 A limit. GPIO17 sources about 6.6 mA to the LED, within the RP4 per-pin limits. The IR sensor and the AS5600 are powered from the RP4's 3.3V pin and draw small currents compared with what that pin can supply. The SN754410NE has internal output clamp diodes for the inductive flyback of the coils; ground pins 4, 5, 12 and 13 on the common rail also conduct heat away from the package. Check the package for heat during long runs. Capacitor C1 (100 µF) across the 12V rail supplies the brief surges of current drawn when the coils switch and keeps VCC2 steady against the voltage spikes that coil switching causes; capacitor C2 (0.1 µF) across the 5V rail bypasses high-frequency switching noise at the SN754410NE logic supply pin.

6.5 IR sensor and RPM measurement

The IR sensor is powered from the Pi's 3.3V pin so its output swings 0 to 3.3 V and can be connected directly to GPIO27 (the Pi inputs are not 5V tolerant). Each time a blade passes through the sensor's detection zone the output produces a pulse; the software polls GPIO27 every 1 ms and treats a low-to-high change as one blade. Because the propeller has two blades, consecutive pulses are half a revolution apart, so RPM = 30 / (seconds between pulses). The pulse spacing is 0.3 s at 100 RPM and 1.5 s at 20 RPM, so a 1 ms poll is ample. The sensor output is HIGH while a blade is in the detection zone, which is the polarity the software expects, and the sensor ground joins the common ground rail.

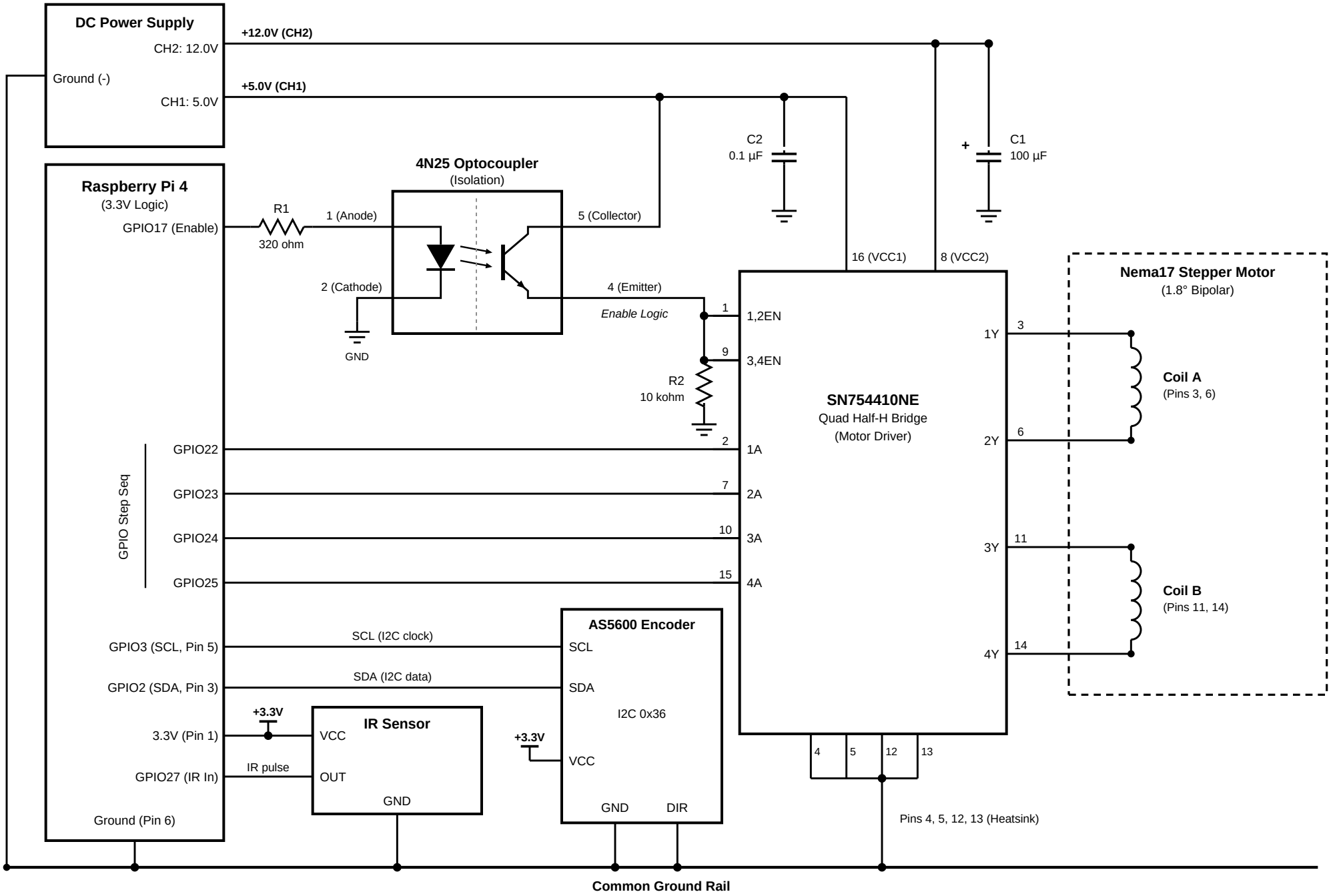
6.6 AS5600 magnetic encoder

The AS5600 senses the direction of the magnetic field of the magnet on the motor shaft and reports it as a 12-bit angle: 4096 counts per revolution, or 0.0879° per count. This is about ten counts per half step (0.9°), so the encoder resolves the shaft position much more finely than the motor steps. The reading is relative to how the magnet sits on the shaft, so the software stores a zero offset when calibrate zero is pressed. The encoder gives a position measurement that does not depend on the step count, which is what lets the software detect missed steps and drive the blade to an angle target.

The encoder is wired as a 3.3V I2C device. Its VCC is on the RP4's 3.3V pin, not 5V, so the SDA and SCL lines stay at 3.3V logic levels that the RP4's I2C pins (GPIO2 and GPIO3, header pins 3 and 5) can accept; 5V on the module would put 5V levels on those pins and could damage them. SDA and SCL are open-drain lines pulled up to 3.3V by the RP4's on-board 1.8 kohm pull-up resistors, so no external pull-ups are fitted. The AS5600 has the fixed I2C address 0x36, and the software reads its angle registers every 2 ms. The DIR pin is tied to ground, which makes the angle increase when the magnet turns clockwise; the software expects this (ENCODER_INVERT_DIR is 0). The OUT and PGO pins are not connected because the angle is read over I2C. The encoder ground is on the common ground rail.

PARMCO Hardware Isolation, Drive, IR Sensor & Encoder Schematic

Subsystem: Motor, Isolation, IR Sensor & Encoder (Checkpoint 3)



GPIO numbers are BCM numbers as used by the class RP4 library. GPIO2 and GPIO3 are used through the Linux I2C driver and are pulled up to 3.3V by the RP4's on-board resistors.
All +3.3V flags are the same net (RP4 3.3V pin). The AS5600 OUT and PGO pins are not connected.